

Introduction to Biometric Technologies and Applications (V)

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Course: Biometric Recognition
DI4025

Outline

Vulnerabilities of Biometrics Systems

Presentation Attacks (PA)

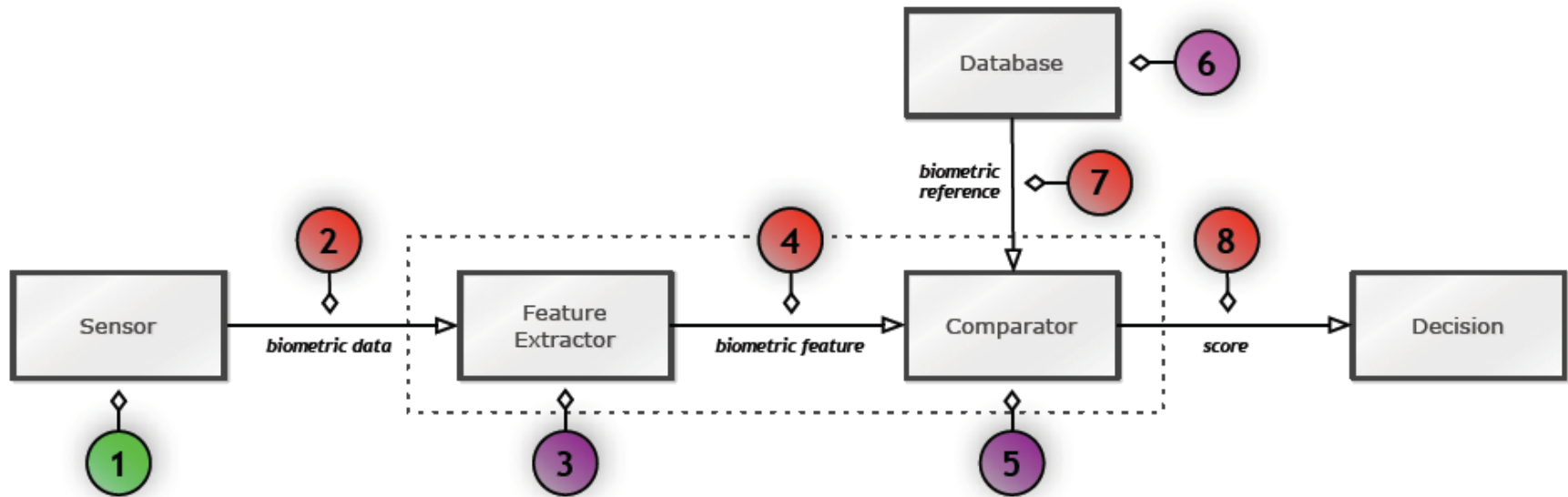
Morphing Attacks (MA)

- Definition, methods, evaluation, solutions

Biometric Security

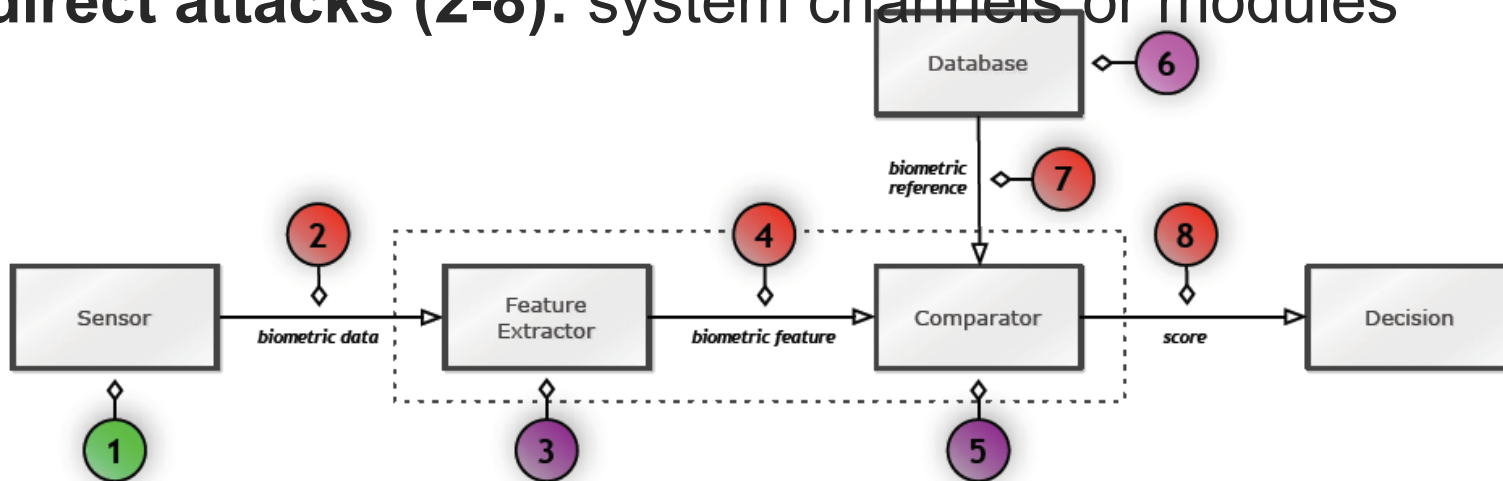
A biometric system is vulnerable to attacks

- Indirect attacks (2-8): system channels or modules
- Presentation (or direct) attacks (1): the capture device



Biometric Security

Indirect attacks (2-8): system channels or modules



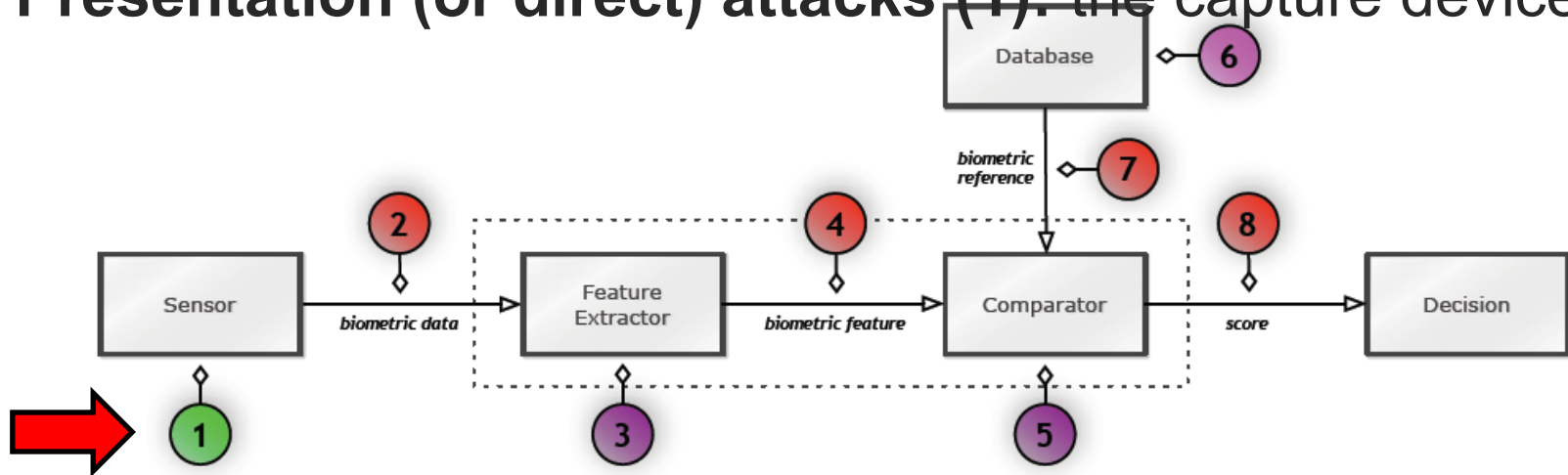
Performed inside the system by:

- bypassing the feature extractor or the comparator (3, 5)
- manipulating the biometric templates in the database (6)
- modifying data in communication channels (2,4,7,8)

Can be counteracted e.g. by intrusion detection or encryption

Biometric Security

Presentation (or direct) attacks (1): the capture device



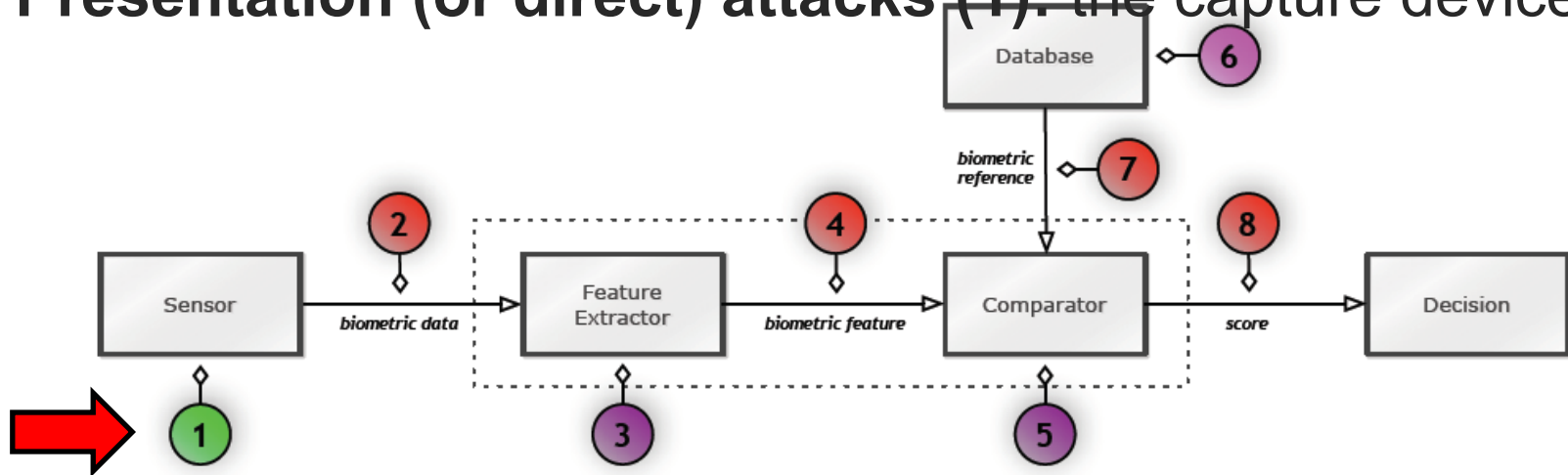
Performed outside the system by:

- fooling the sensor by presenting a **fake** biometrics sample



Biometric Security

Presentation (or direct) attacks (1): the capture device



Performed outside the system by:

- fooling the sensor by presenting a **fake** biometrics sample

The attack is done outside of the physical limits of the system

The sensor itself is not replaced nor tampered.

Knowledge of the inners of the system is not required, no special skills about computer science either

(In this lecture we are *mostly* concerned with **these attacks**)

Presentation Attacks: Categories

Impersonation

- to be somebody else
(positive access 1:1 or 1:N)

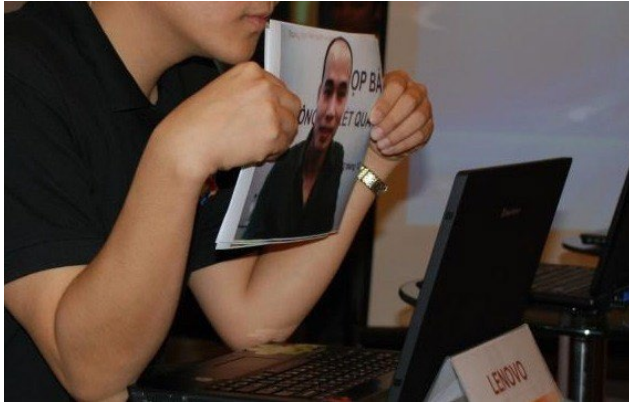
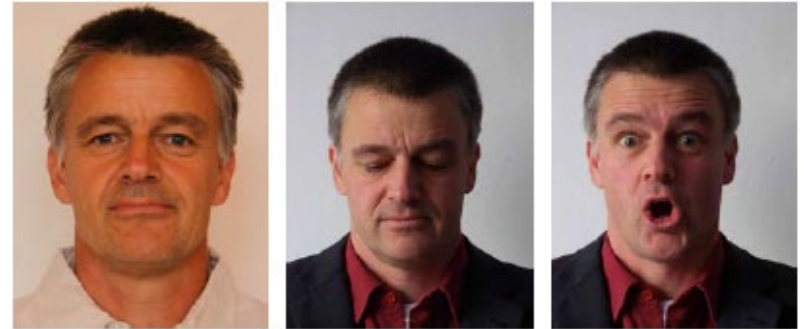


Image Source: <http://upshout.net/game-of-thrones-make-up>

Concealment

- to evade recognition
(negative 1:N identification)



- to evade face detection

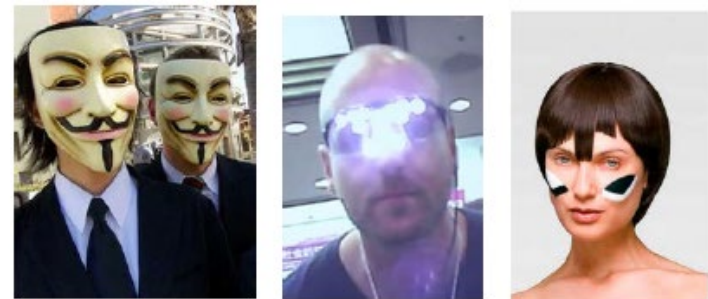


Image Source: <https://www.youtube.com/watch?v=LRj8whKmN1M>

Image Source: <https://cvdazzle.com>

Presentation Attacks: Definitions

Presentation Attack (PA)

- presentation to the biometric capture subsystem with the goal of interfering with the operation

Presentation Attack Detection (PAD)

- automated determination of a presentation attack

(ISO/IEC 30107 PAD - Part 1: Framework)

Impostor

- subversive biometric capture subject who attempts to being matched to someone else's biometric reference

Identity concealer

- subversive biometric capture subject who attempts to avoid being matched to their own biometric reference

(ISO/IEC 2382-37: Vocabulary)

Presentation Attacks (PA): Examples

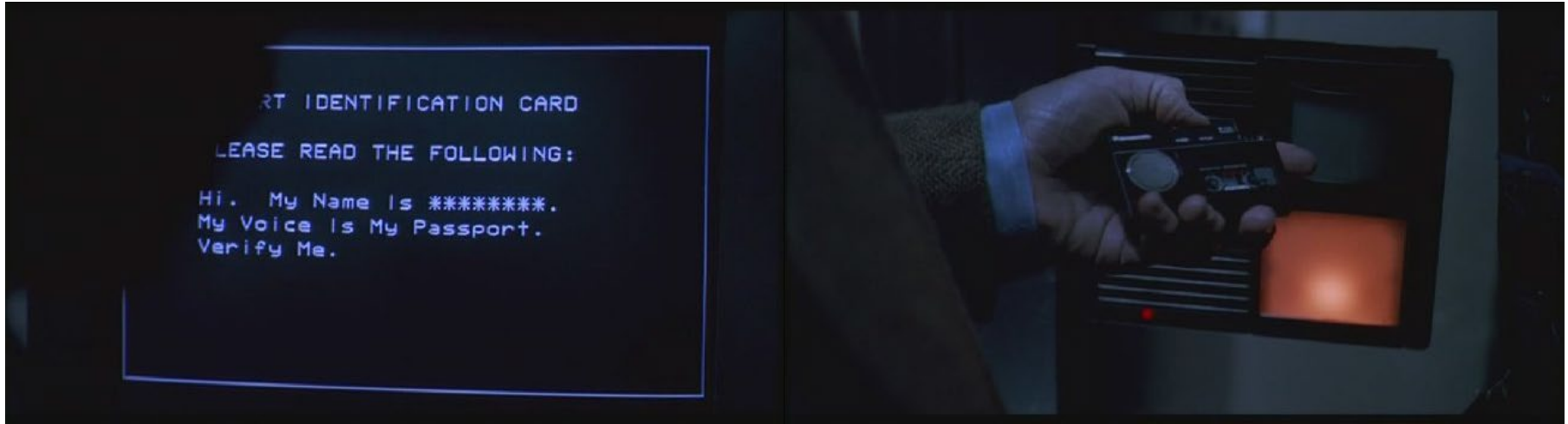
Presentation Attacks in Movies

James Bond: Diamonds Are Forever (1971)



Presentation Attacks in Movies

Sneakers (1992)



Replay a voice recording in front of a speaker recognition system!

Presentation Attacks in Movies

Demolition Man (1993)



Present an eyeball in front of a iris scanner !

Presentation Attacks in Reality

Bank robbery (2010)



Conrad Zdzierak used a silicon masks to pass himself off as a black character "SPFX The Player" during robberies!

Presentation Attacks in Reality

Hong Kong - Vancouver (Jan 2011)



A passenger boarded a plane in Hong Kong with an old man mask and arrived in Canada!

Presentation Attacks in Reality

iPhone 5s - Touch ID (Sep 20 2013)



How many days will it take to spoof it? **2 days !**

Spoofed by the Chaos Computer Club (1st public ...)

<https://www.theverge.com/2016/5/2/11540962/iphone-samsung-fingerprint-duplicate-hack-security>

Presentation Attacks in Reality

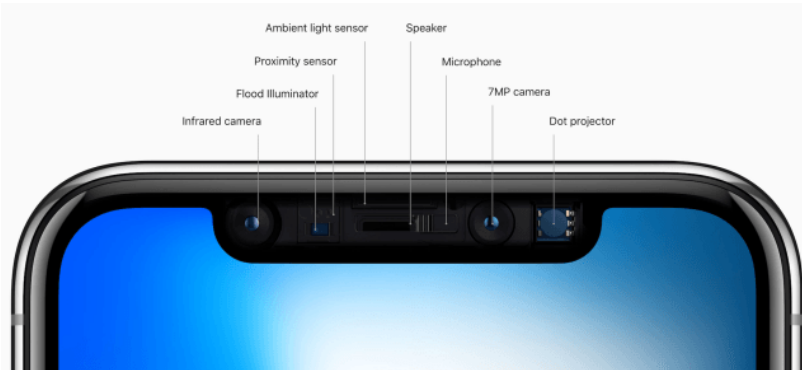
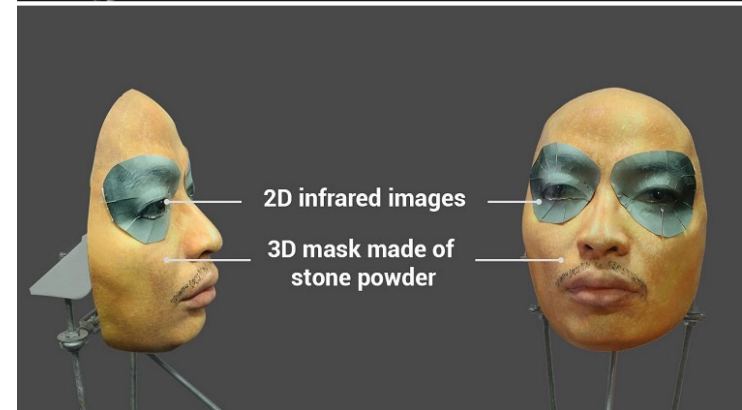
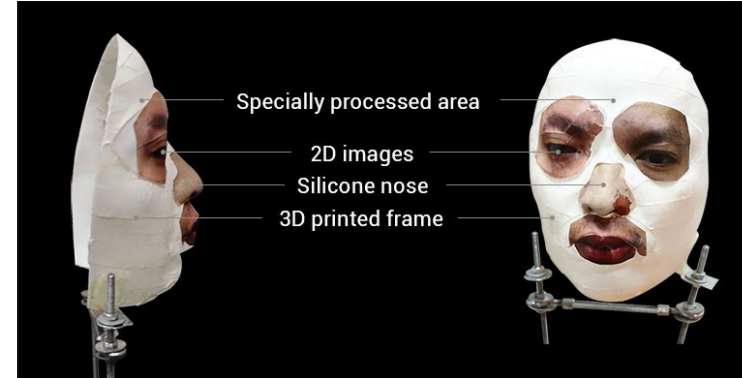
Samsung Galaxy S8 Iris spoofed by CCC (May 23 2017)



<https://media.ccc.de/v/biometrie-s8-iris-en>

Presentation Attacks in Reality

iPhone X FaceID robust to masks (Sep 2017)



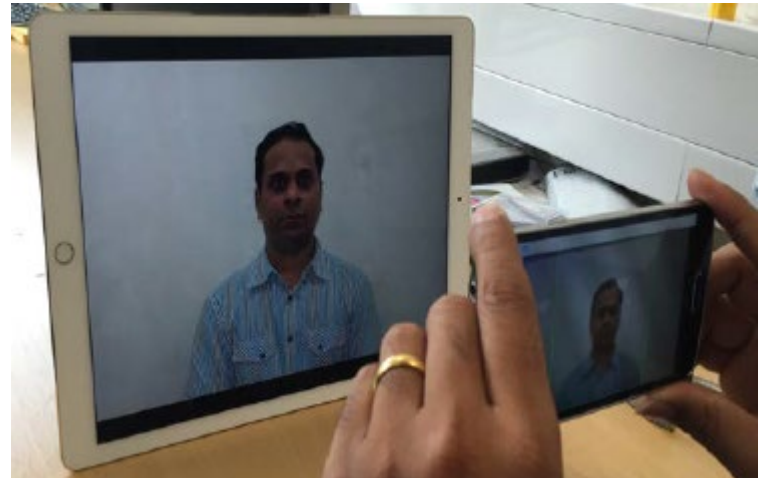
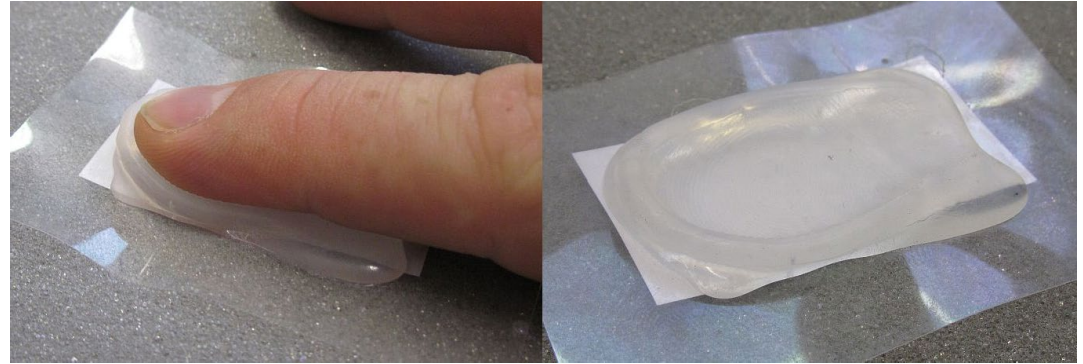
Spoofed by Bkav (Nov 27 2017), www.bkav.com

Presentation Attacks (PA): Methods

Presentation Attacks: Methods

A **fake biometric** characteristic is presented to the sensor

- artefact, replica, mutilations, digital replay, . . .

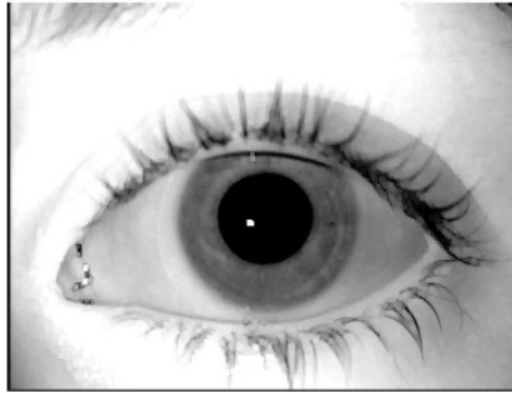


Presentation Attacks: iris

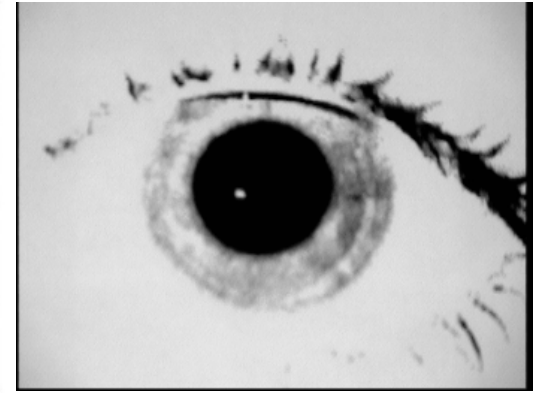


Capturing a fake iris

real



fake



Cosmetic contact lens



“Contact lens”

Presentation Attacks: fingerprints



(a)



(b)



(c)



(d)



(e)



(f)



(g)



(h)

**Obtaining a fake
fingerprint
(with cooperation)**

Presentation Attacks: fingerprints



(a)



(b)



(c)



(d)



(e)



(f)



(g)



(h)

**Obtaining a fake
fingerprint
(without
cooperation)**

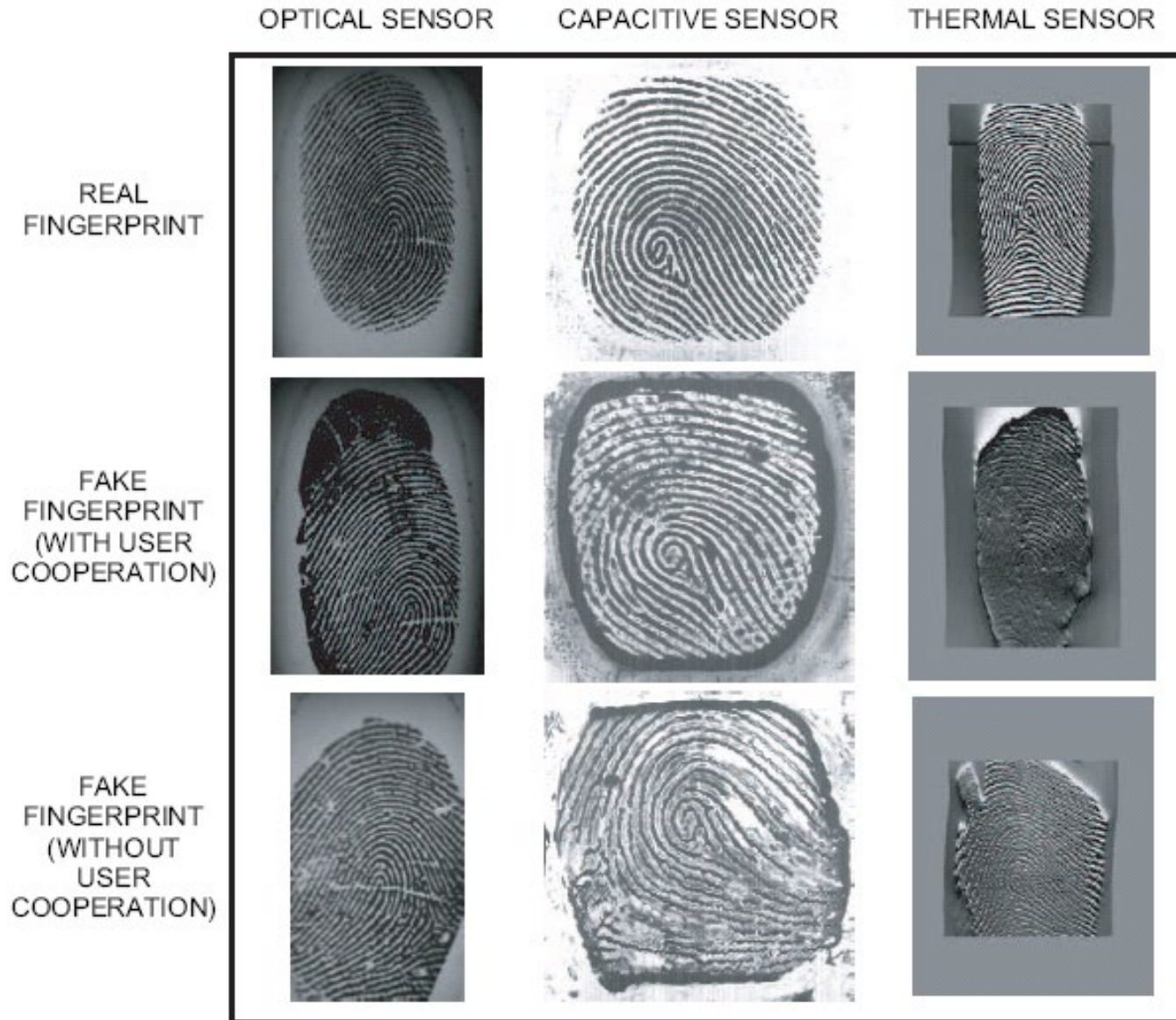
Presentation Attacks: fingerprints

**Obtaining a fake
fingerprint
(without
cooperation)**



Source: <https://www.ccc.de/en/tags/apple>, (2013)

Presentation Attacks: fingerprints



PA: fingerprints (concealment)

Intentional alteration to avoid identification (e.g. border crossing)



(a)

(b)

(c)

(d)

Fig. 1. Photographs of altered fingerprints. (a) Transplanted friction ridge skin from sole [6]. (b) Fingers that have been bitten [7]. (c) Fingers burnt by acid [8]. (d) Stitched fingers [9].



(a)



(b)

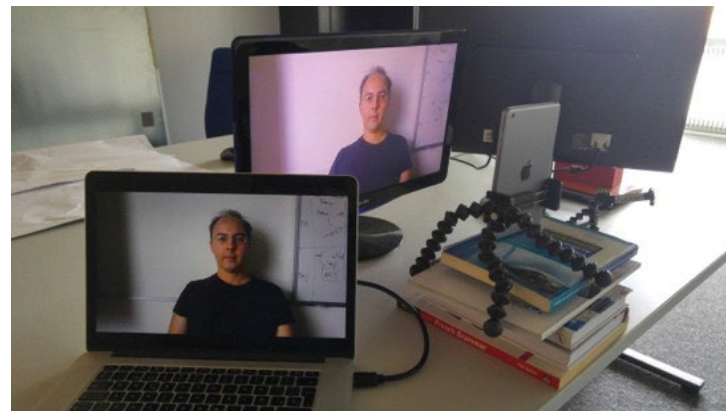
Fig. 3. Inked impressions before and after fingerprint alteration (a) of Gus Winkler [5] (pattern type is altered from double loop to left loop) and (b) of Jose Izquierdo [15] (altered by switching two parts of a "Z" shaped cut on the fingertip).

Presentation Attacks: faces

Printed photo



Photo/video screen



Rigid mask



Cost ~300 USD

Non-rigid mask



Cost >3000 USD

<http://www.thatmyface.com>

Presentation Attacks: faces

Which one is real or fake?



Presentation Attacks: faces

Which one is real or fake?



All are fakes: print (left), iPhone (middle) and iPad (right)

Presentation Attacks: faces (concealment)

Face disguise to avoid identification (e.g. organized crime)

<https://www.dailymail.co.uk/news/article-2153346/Black-armed-robber-disguised-white-man-using-latex-mask.html>

The man in the latex mask: **BLACK** serial armed robber disguised himself as a **WHITE** man to rob betting shops

- Henley Stephenson wore the disguise during a 12-year campaign of hold-ups at betting shops and other stores across London
- He was part of a three-man gang jailed for a total of 28 years
- CCTV footage showed him firing a semi-automatic pistol into the ceiling during a raid on a betting shop
- The mask was bought from the same London shop which supplied masks used in the £40m Graff Diamonds heist

By ROB PREECE and REBECCA CAMBER FOR THE DAILY MAIL

PUBLISHED: 18:22 BST, 1 June 2012 | UPDATED: 17:21 BST, 2 June 2012

Most masked robbers opt for a balaclava to hide their identity.

Not this one. Henley Stephenson, 41, eluded police for more than ten years thanks to an extraordinarily lifelike latex mask, which turned him into a white skinhead.

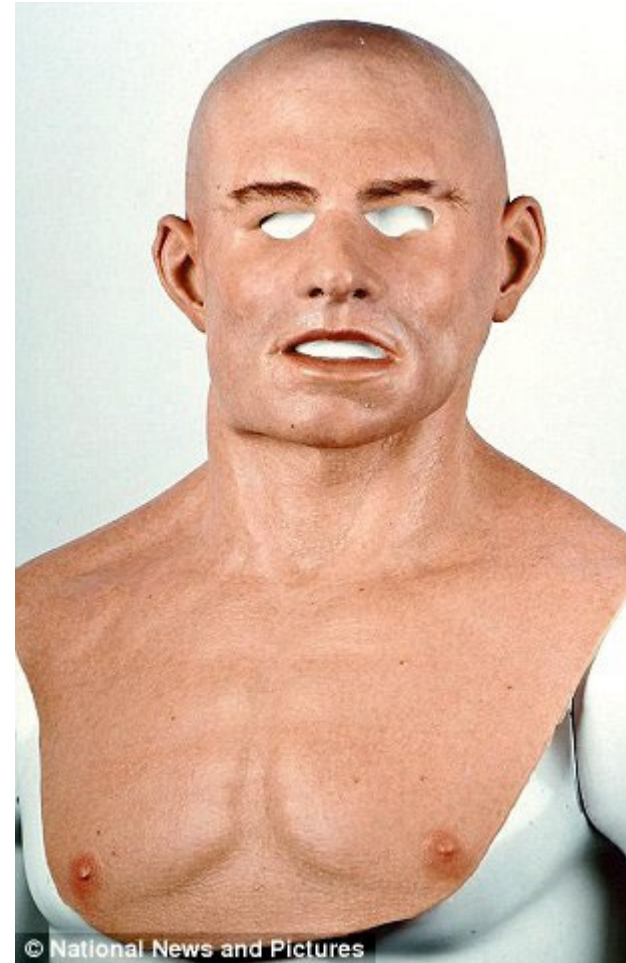
Officers discovered that their man was in fact black when they finally caught up with Stephenson after a string of armed raids dating back to 1999.



© National News and Pictures



© National News and Pictures



© National News and Pictures

Presentation Attacks: faces (concealment)

“Make-Up attack”



(a) before

(b) after

(c) target

Image Source: <http://www.antitza.com/makeup-datasets.html>

Presentation Attacks: Solutions?

Biometric Security

Presentation attack is a major threat

because it can be created and performed by anyone with no specific skills in computer science



PAD/spoof detection critical, especially if authentication is **unsupervised / not monitored** (e.g. smartphone)

Important **public perception**, which can derail the success of a biometric system if not addressed properly

Presentation Attack Detection: solutions?

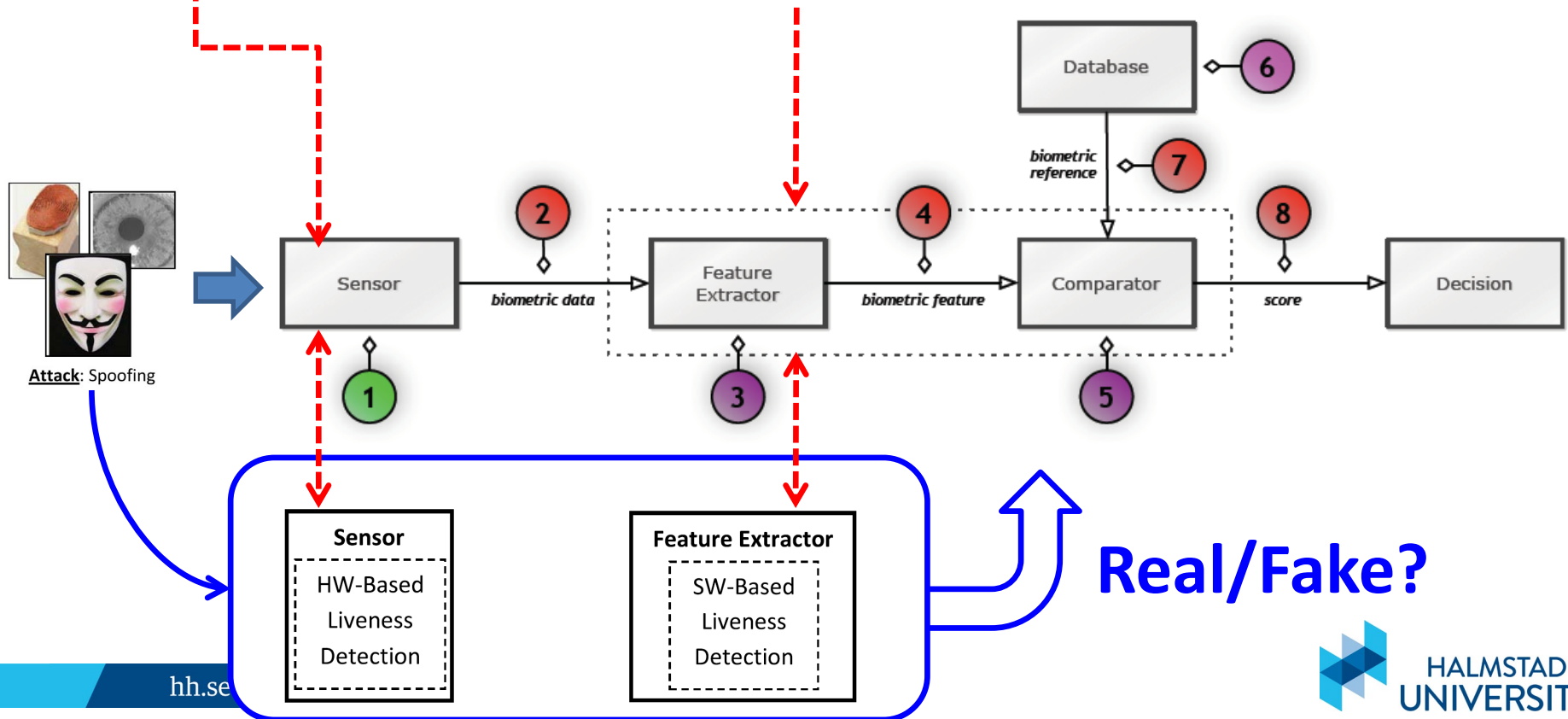
Two possibilities:

Hardware device attached to the sensor (mainly for physiological properties)

- Added cost, extra size, more intrusive

Software once the sample is acquired (action or challenge/response)

- Less expensive, less intrusive, more time-consuming



Presentation Attack Detection: solutions?

Countermeasures: check for signs of **human vitality**
(also known as "**aliveness detection**")

Physiological properties

- blood pulse
- perspiration
- eye blinking
- skin/tissues properties...

(Un-)voluntary actions

- blinking
- fluctuations in pupil size
- pupil/eye/head/body movement

Challenge-response (~captchas)

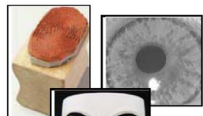
- smile, blink
- say a random number (voice, face)...

... in **real time**, with **high accuracy**, without increasing **costs** too much, without **inconvenience** to legitimate users

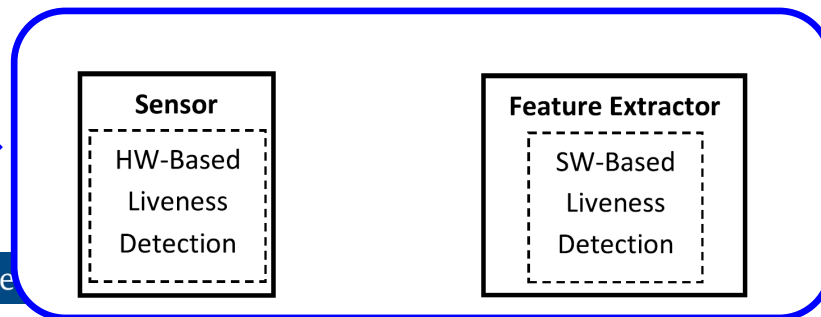
PAD: performance evaluation

Four possible outputs, two types of errors

		Output of the system	
		Real	Fake
Claim	Real	OK	Bona Fide Presentation Classification Error (BPCE) • % of bona fide classified as PA • Real users annoyed
	Fake	Attack Presentation Classification Error (APCE)  • % of PA classified as bona fide • Fake users get illegitimate use	OK



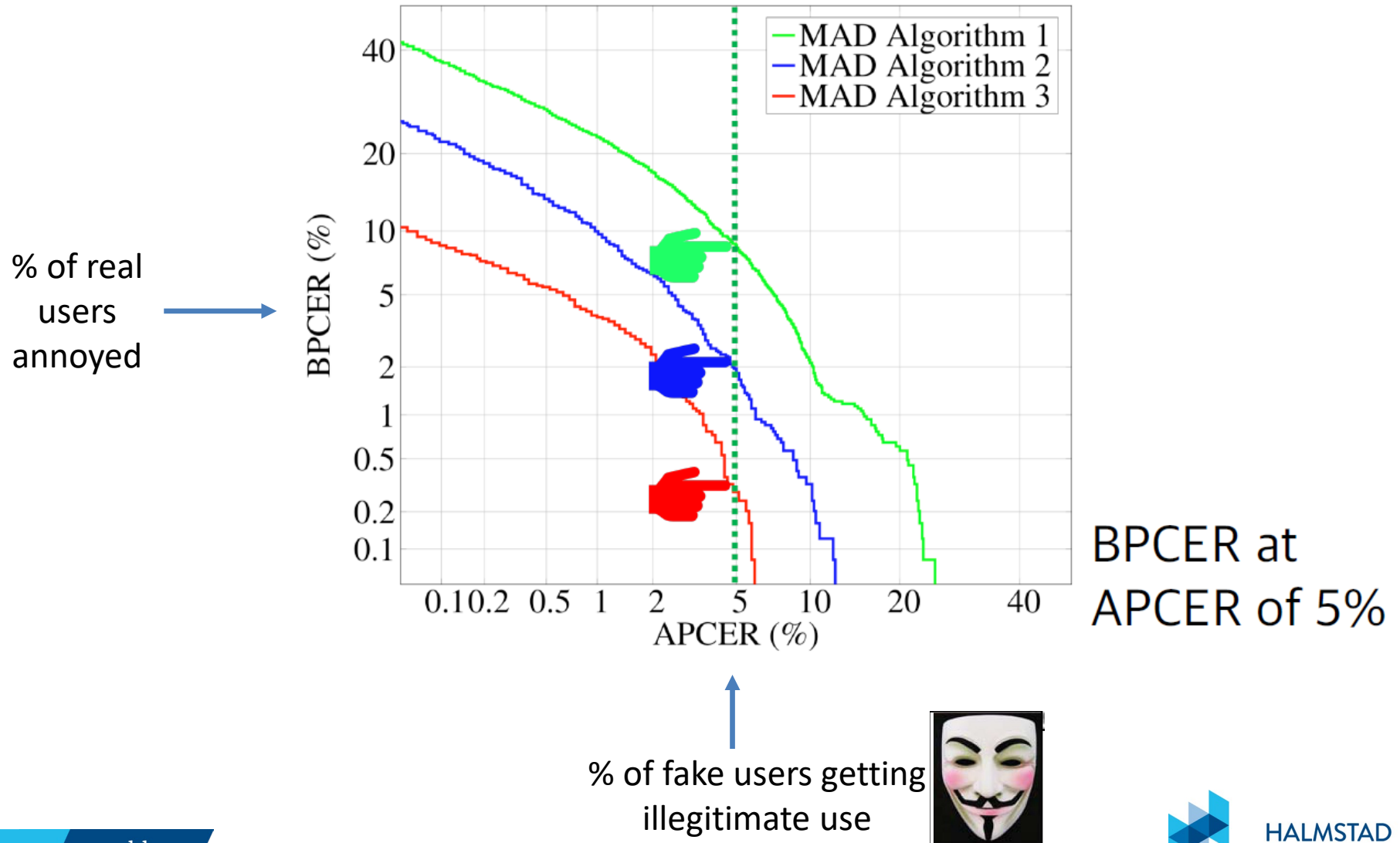
Attack: Spoofing



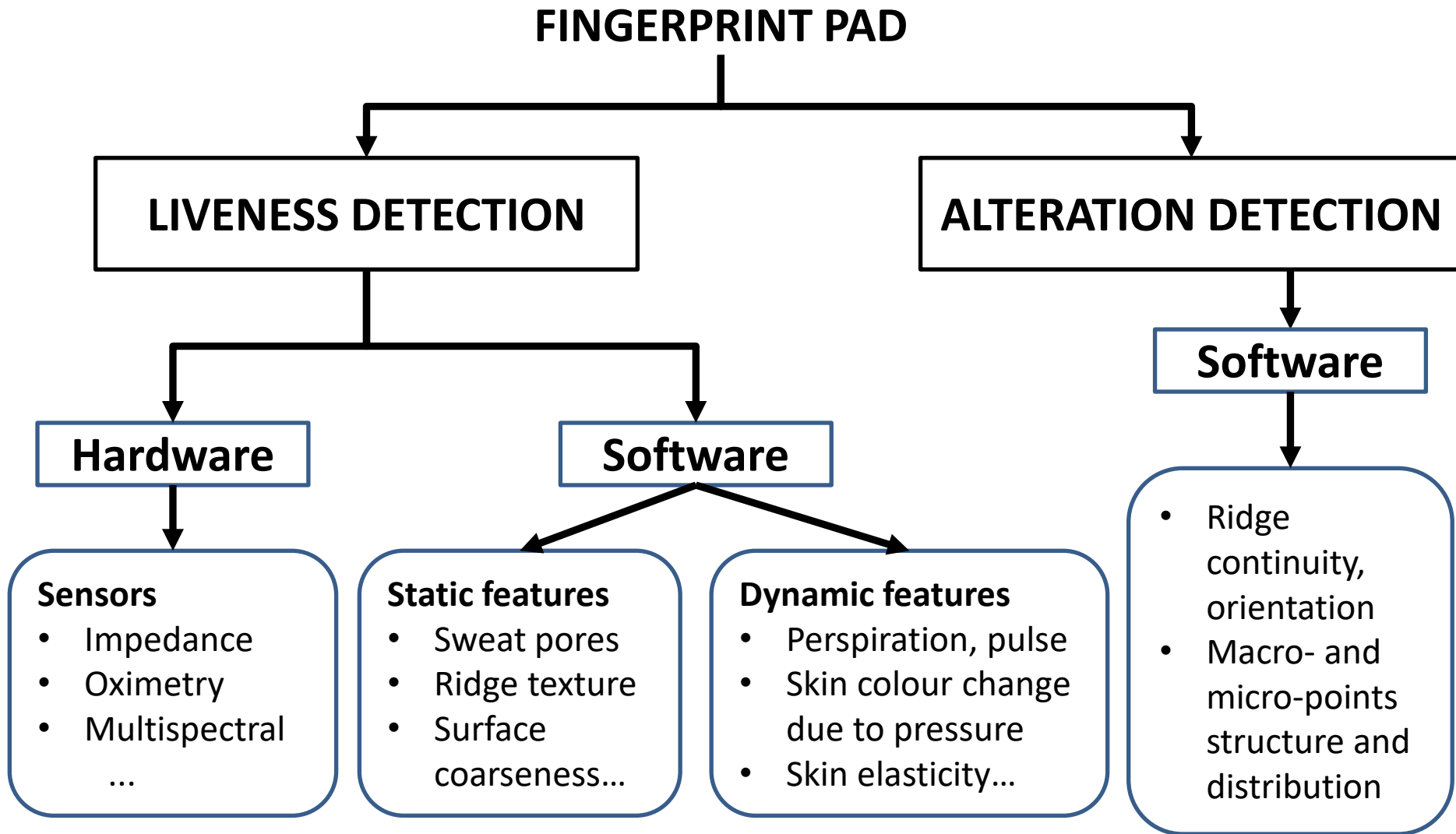
**Real (bona fide)
or Fake?**

PAD: performance evaluation

Performance Evaluation for MAD Algorithms



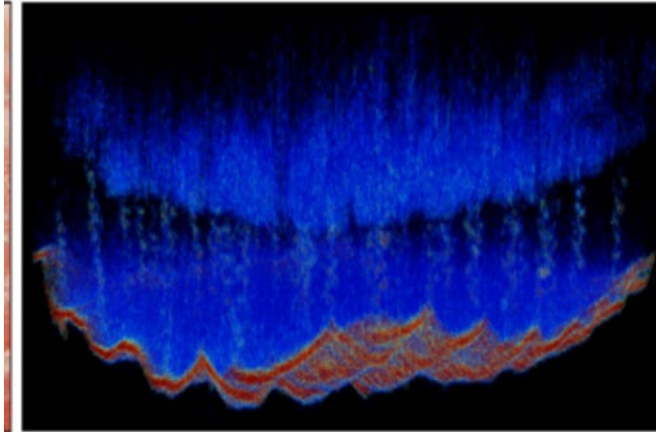
Presentation Attack Detection: fingerprint



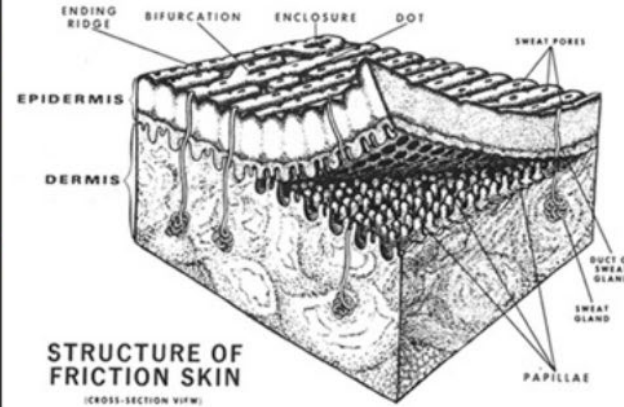
PAD: fingerprints (examples)

Sweat pores and glands appearing in OCT scans

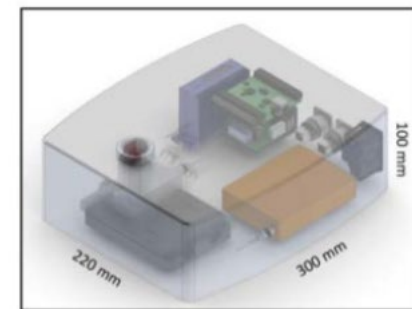
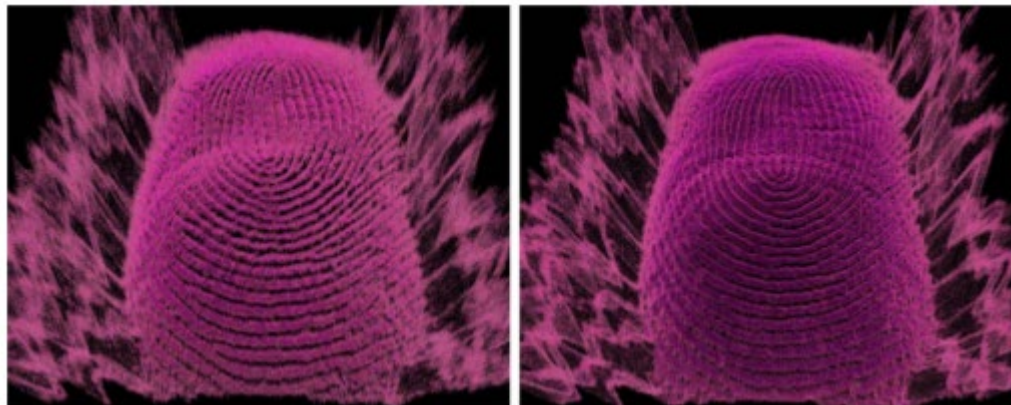
Detection of outer and inner layers



b



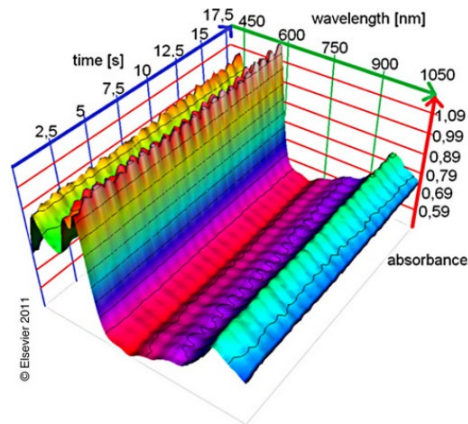
c



Source: BSI

PAD: fingerprints (examples)

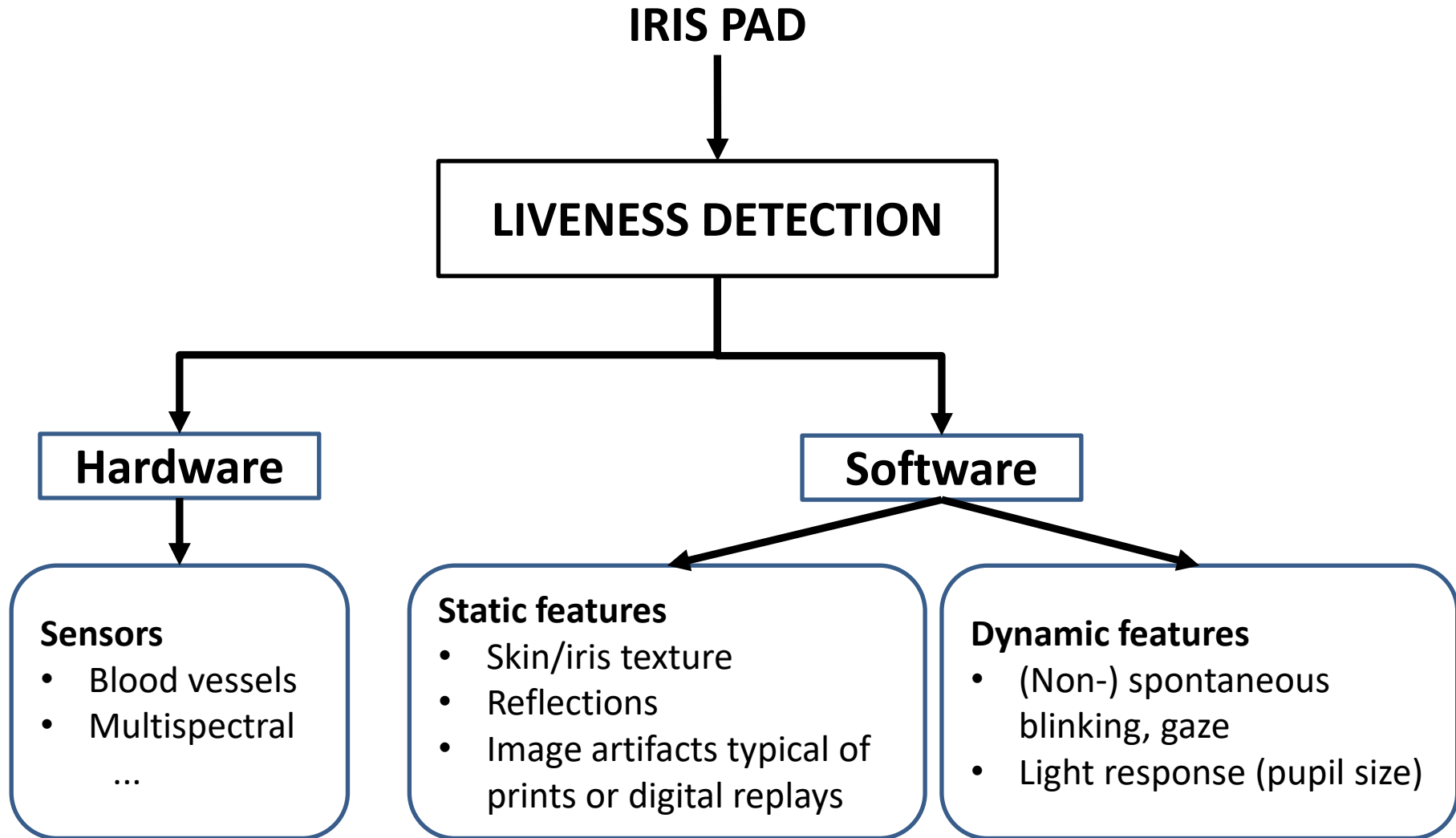
Change of light absorption due to blood circulation



Time-series of a fingerprint scan



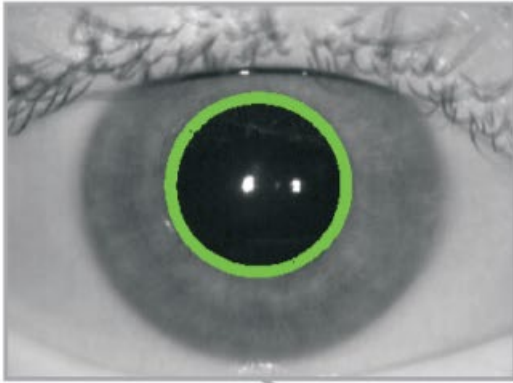
Presentation Attack Detection: iris



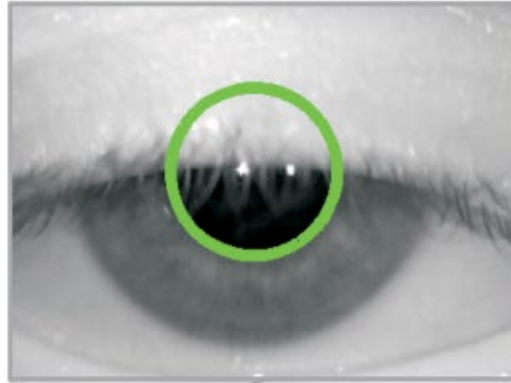
PAD: iris (examples)

Eye blinking & pupil response to light stimulus

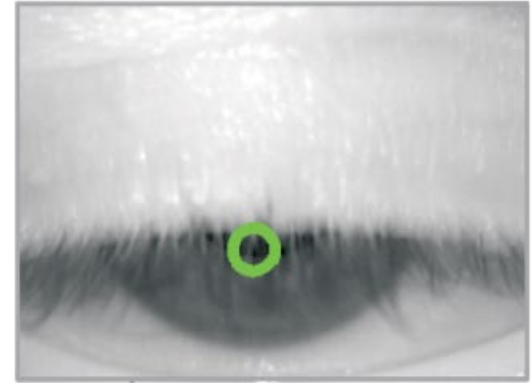
correct segmentation
of high quality image



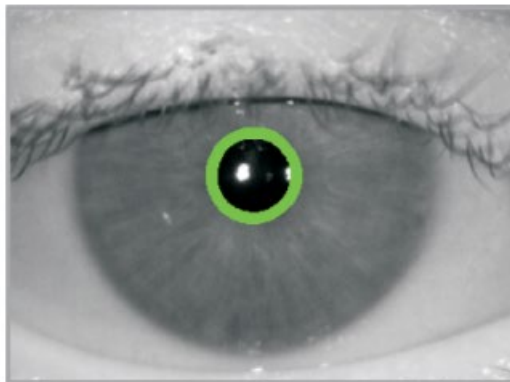
correct segmentation
of low quality image



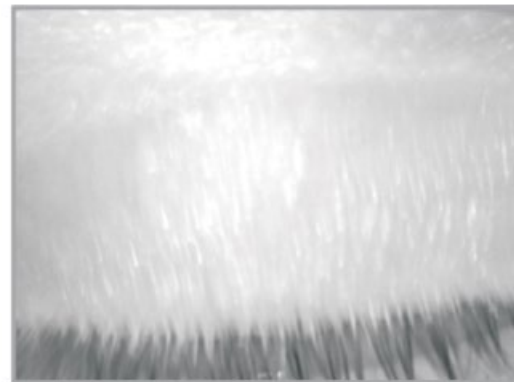
incorrect segmentation
when blinking



correct segmentation
after light stimulus



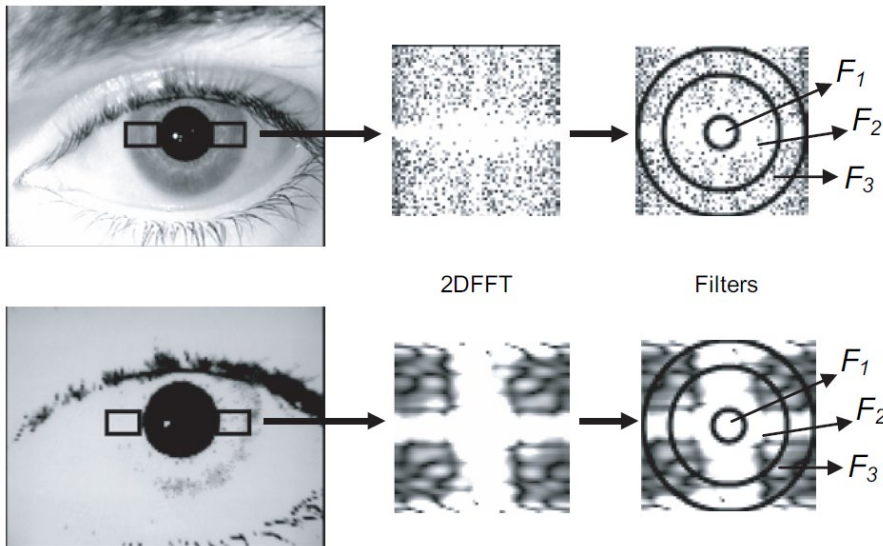
no pupil detected (correct)
when blinking



PAD: iris (examples)

Image artifacts typical of prints or digital replays

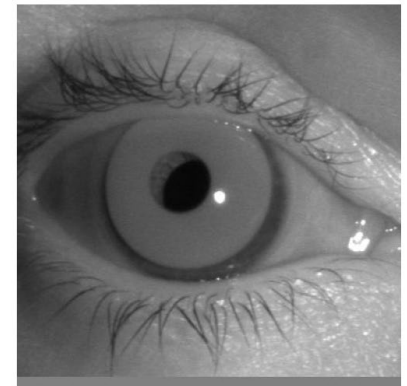
Paper print



Contact lens

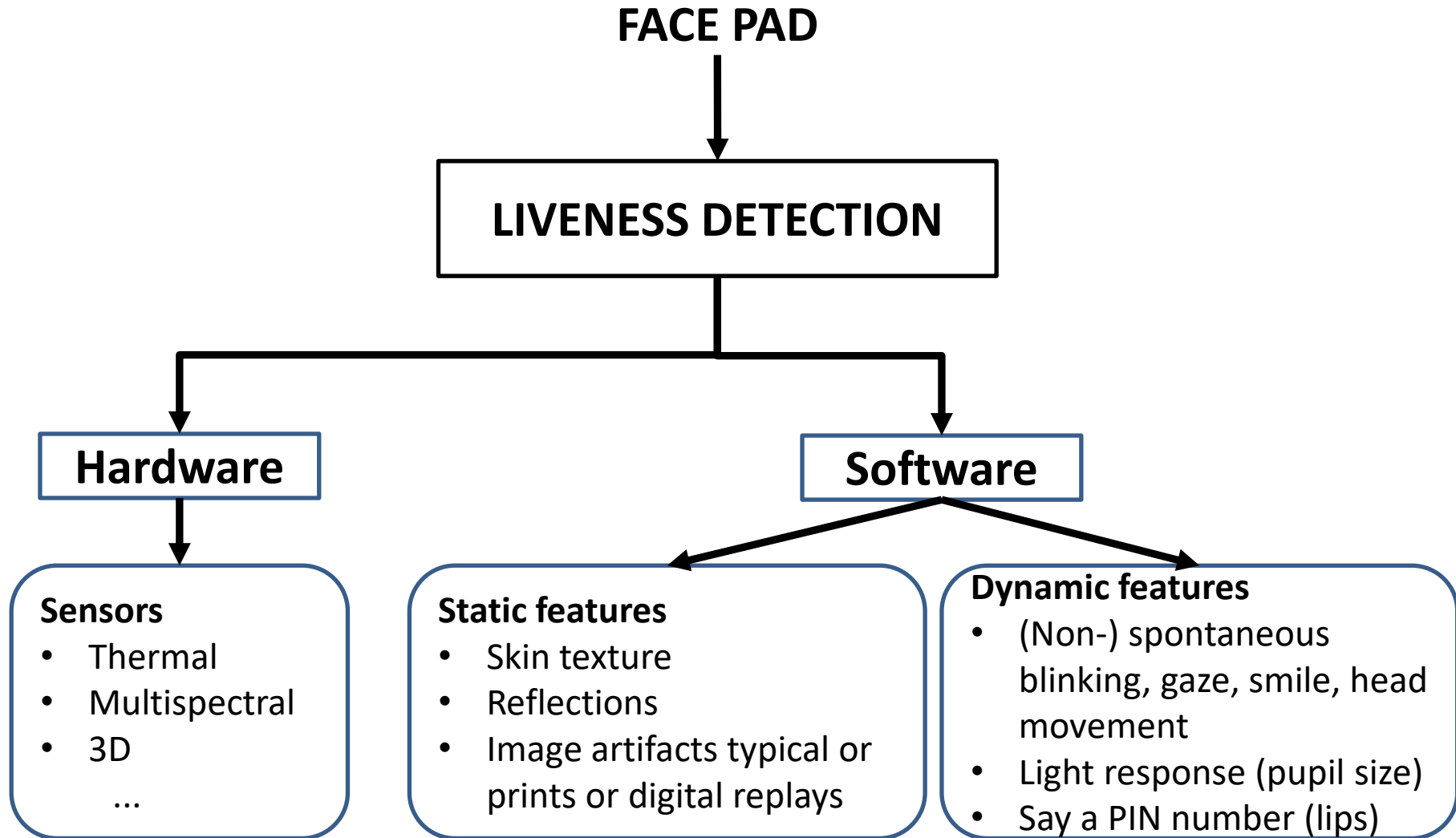


under visible light



under infrared

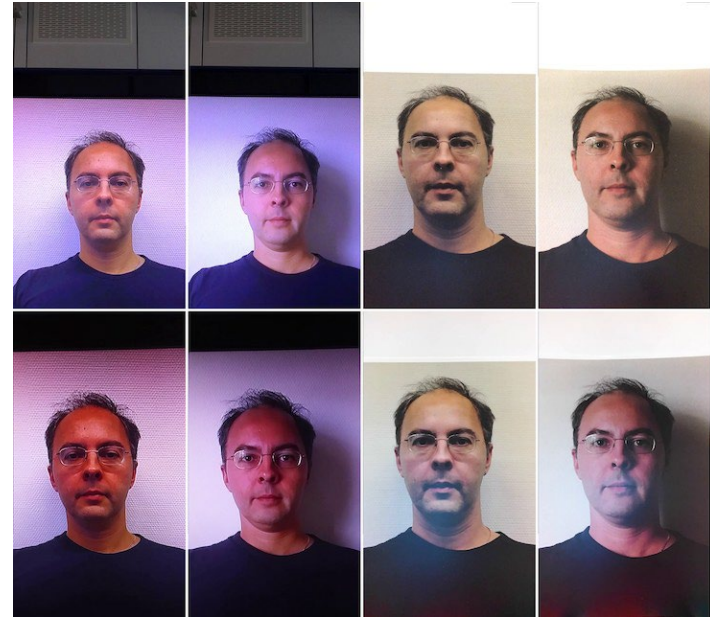
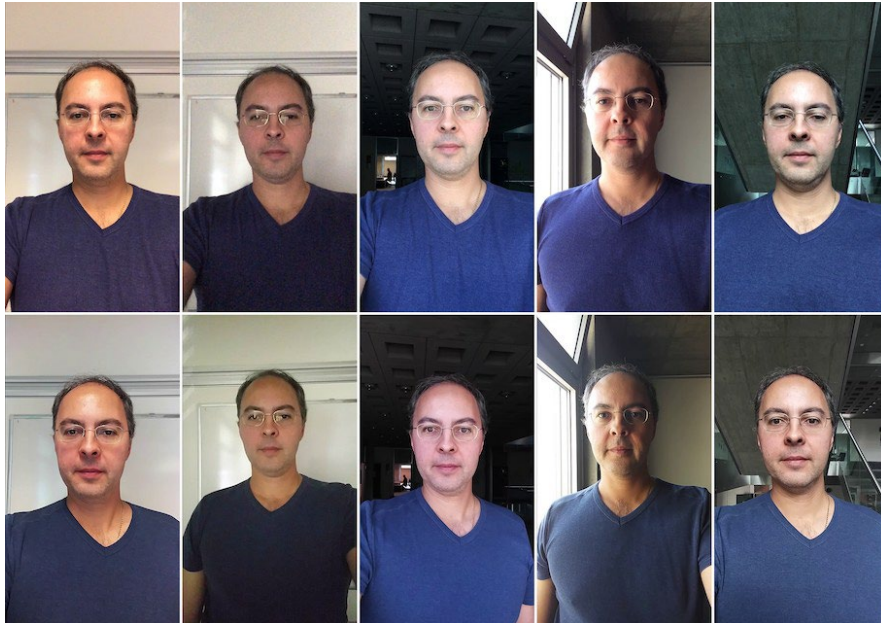
Presentation Attack Detection: face



PAD: face (examples)

Printed paper or photo/video screen replay

- Printer halftoning artifacts
- Reduced texture and color diversity due to printer/screen dynamic range
- No motion (e.g. blinks) in paper, no challenge-response



PAD: face (examples)

Example printed paper or masks

- Differences between visible and near-infrared

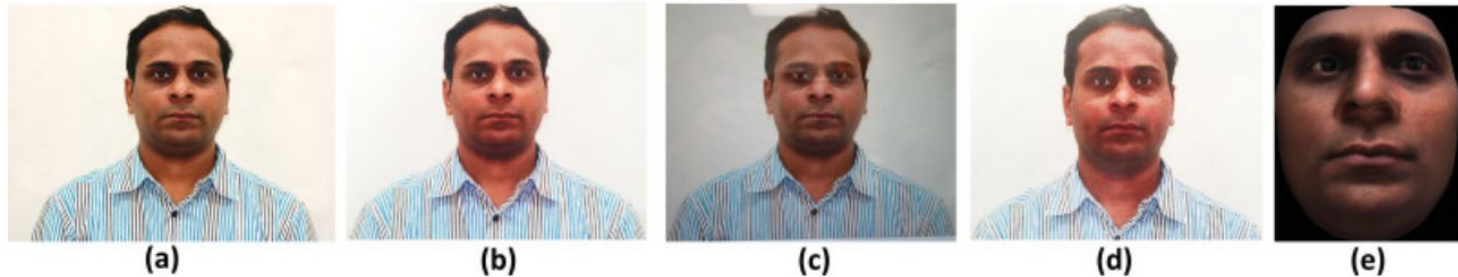


Fig. 2. (a) Bona fide facial image and examples of face artifacts: (b) laser print face artifact; (c) display face photo artifact using an iPad; (d) inkjet print face artifact; (e) 3D face mask.

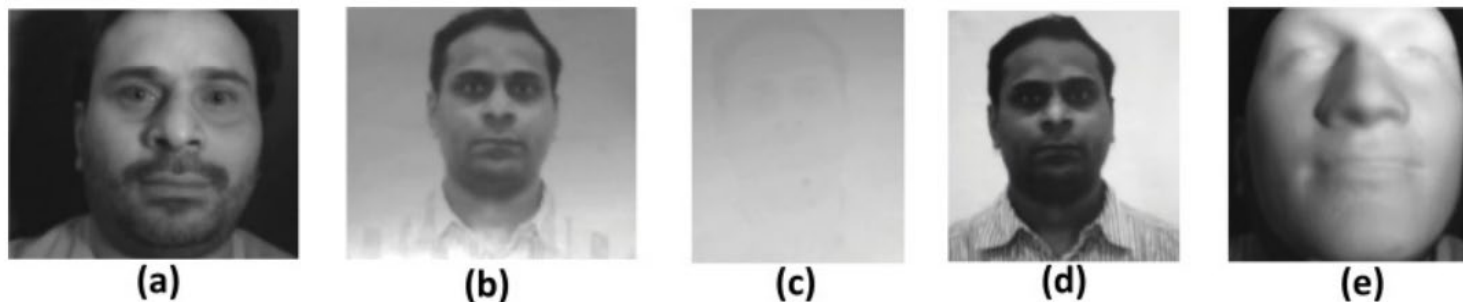
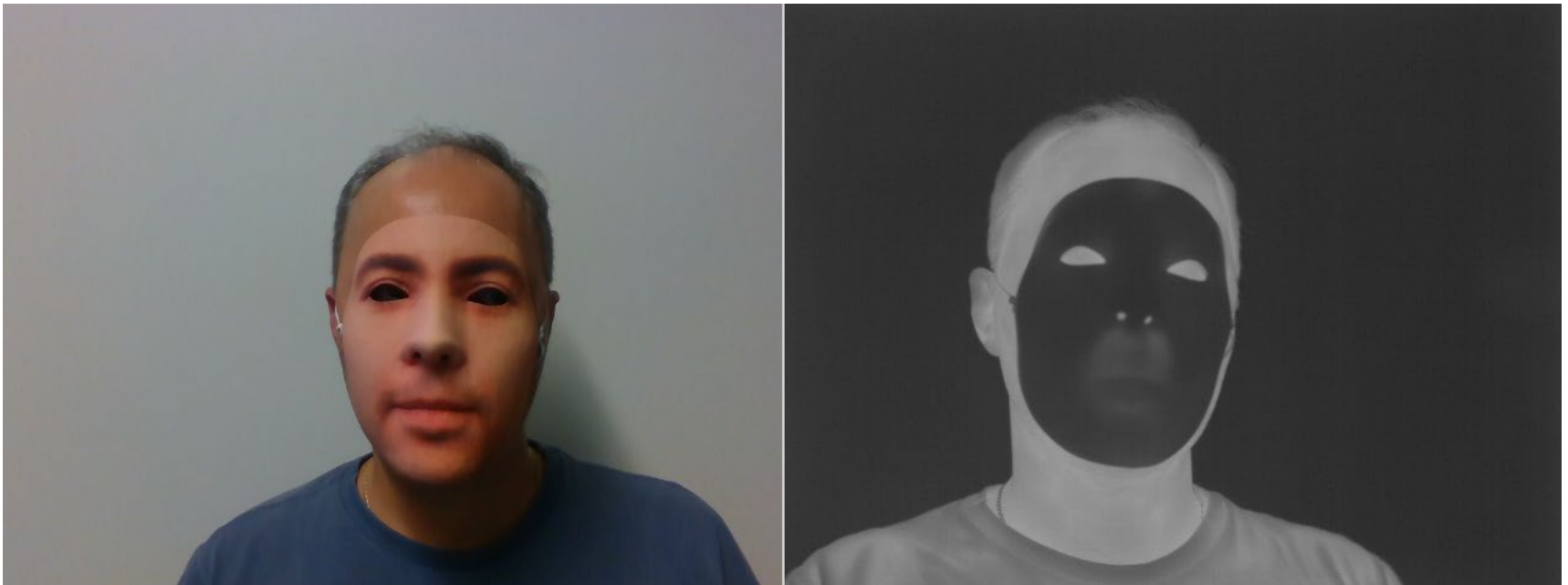


Fig. 8. Illustration of near-infrared face capture of (a) bona fide (real) face; (b) photo print using a laser printer; (c) photo print using an inkjet printer; (d) display attack using an iPad; and (e) 3D mask attack.

PAD: face (examples)

Rigid mask

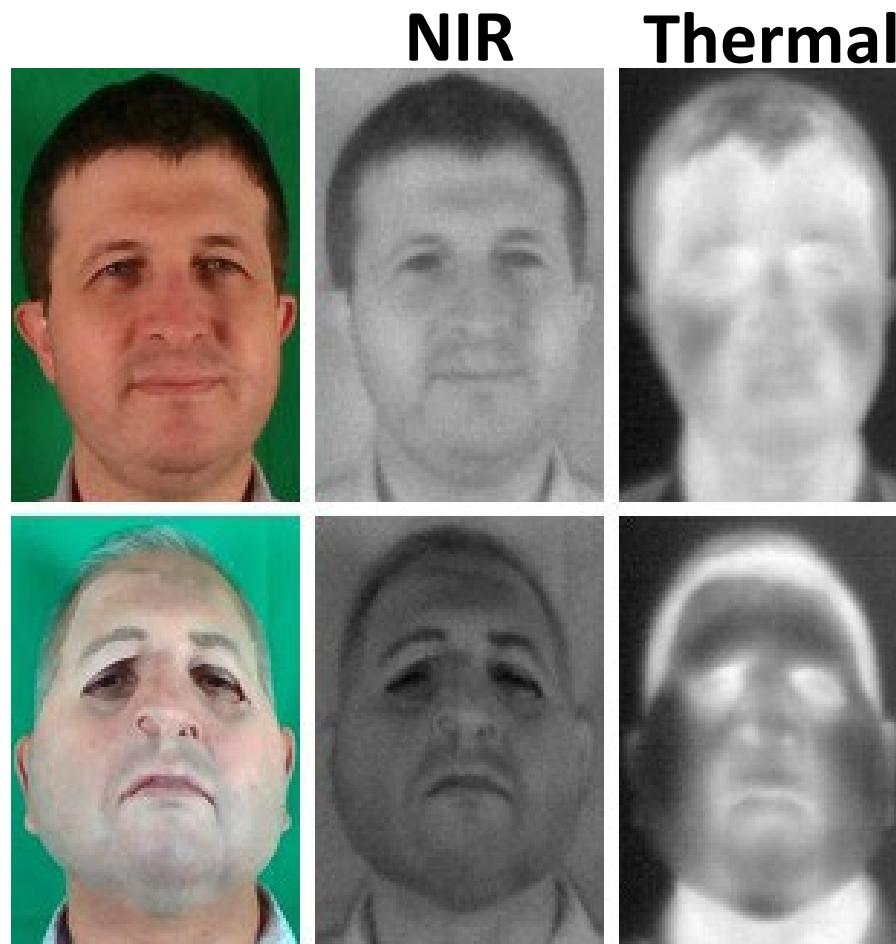
- No facial motion (may have eye motion if it has holes)
- No texture in near-infrared (NIR)
- Not much heat absorption (evident in thermal imaging)



PAD: face (examples)

Non-rigid mask

- Facial motion is possible
- Texture appears in NIR & silicone absorbs heat!



PAD: face (examples)

Example eye-blink motion

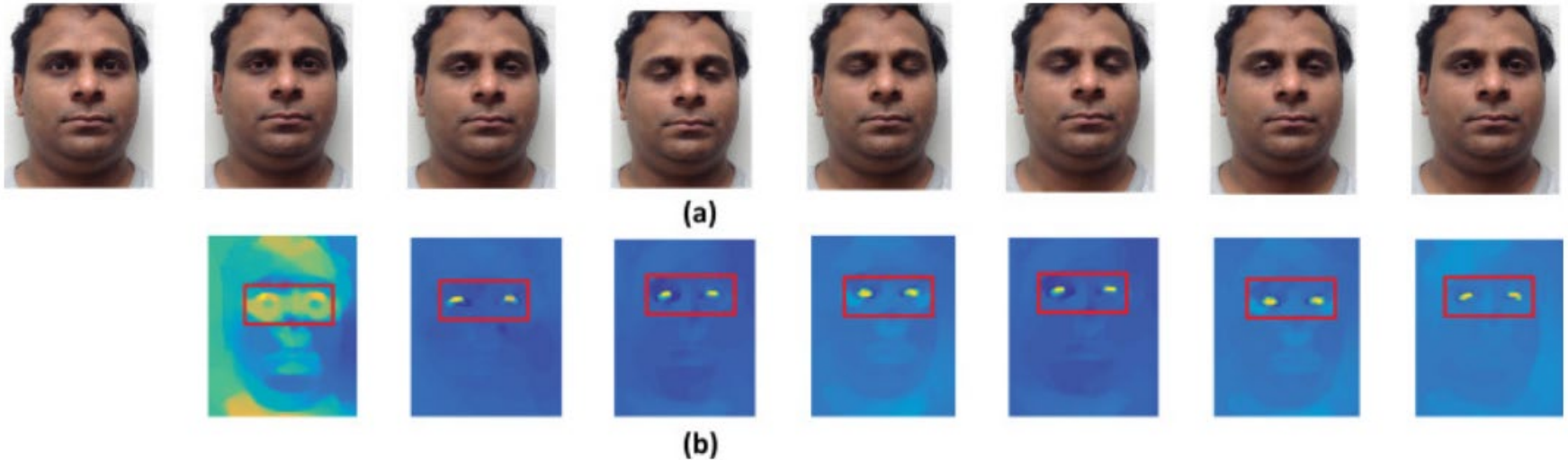
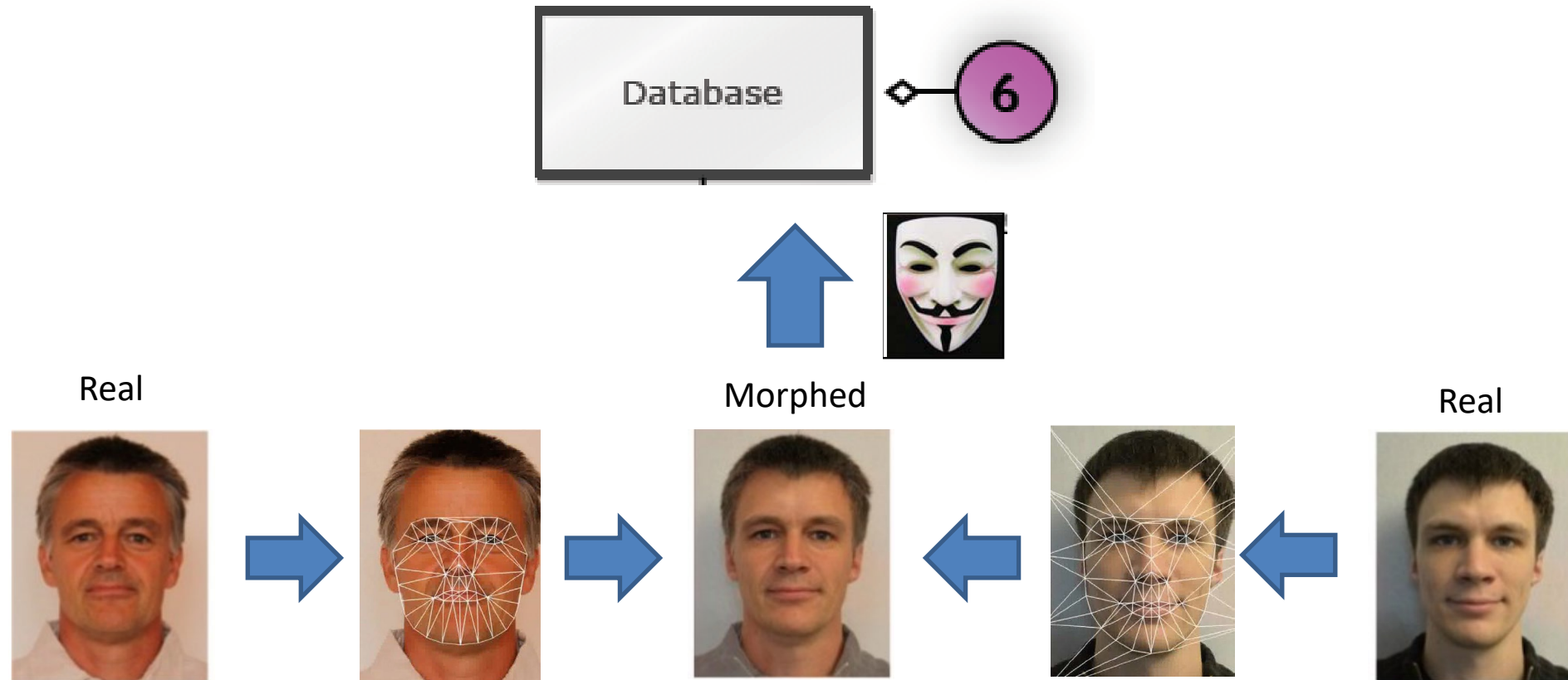


Fig. 9. Illustration of eye-blink detection as a face presentation attack detection mechanism: (a) video frames and (b) corresponding optical flow.

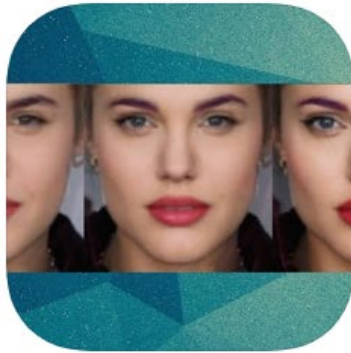
Morphing Attacks (MA)

Morphing Attacks: Definition

Morphing Attacks (MA): a morphed image (50/50 average of two people's faces) is used for enrolment



Morphing Attacks: Plenty of apps



FaceFilm - Baby Maker & Aging 4+

future baby generator app

Jiangguo Yan

★★★★★ 4.5, 3K Ratings

Free · Offers In-App Purchases

Screenshots iPhone iPad



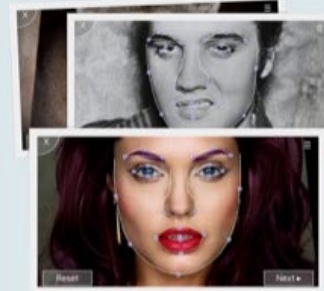
Blend two face photos into one, and see what you get.



See how face changed in years with your favorite song.

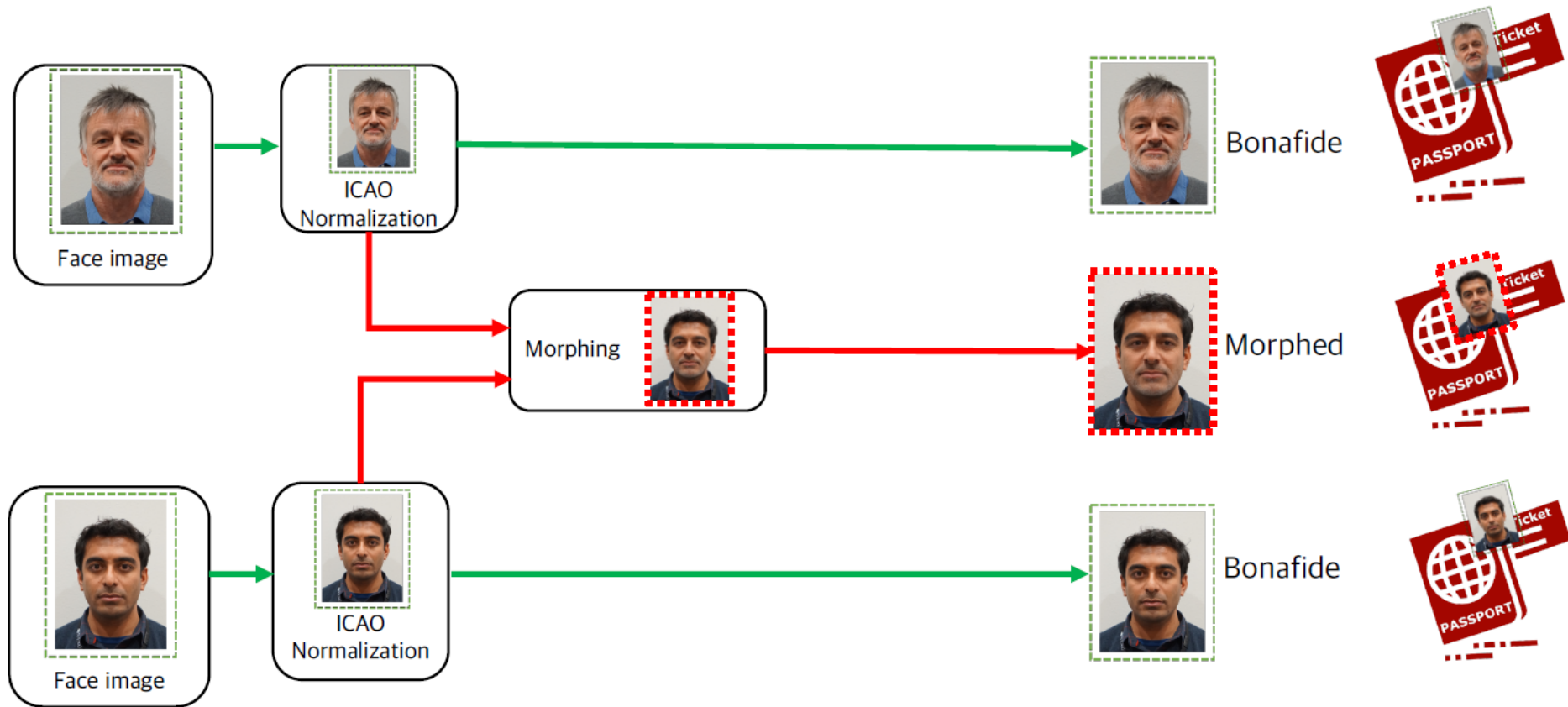


Transitions are really smooth in saved video.

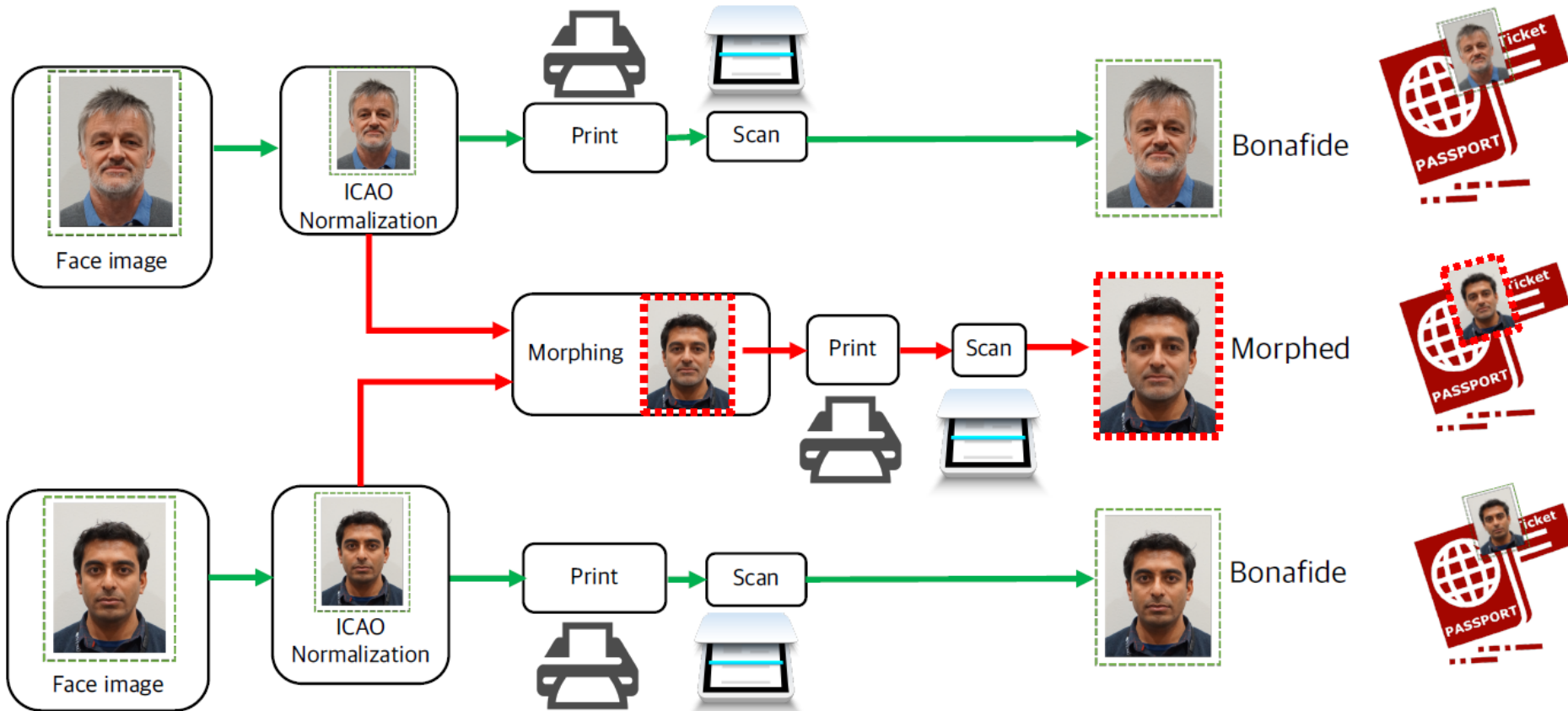


Edit face photos with computer vision algorithms

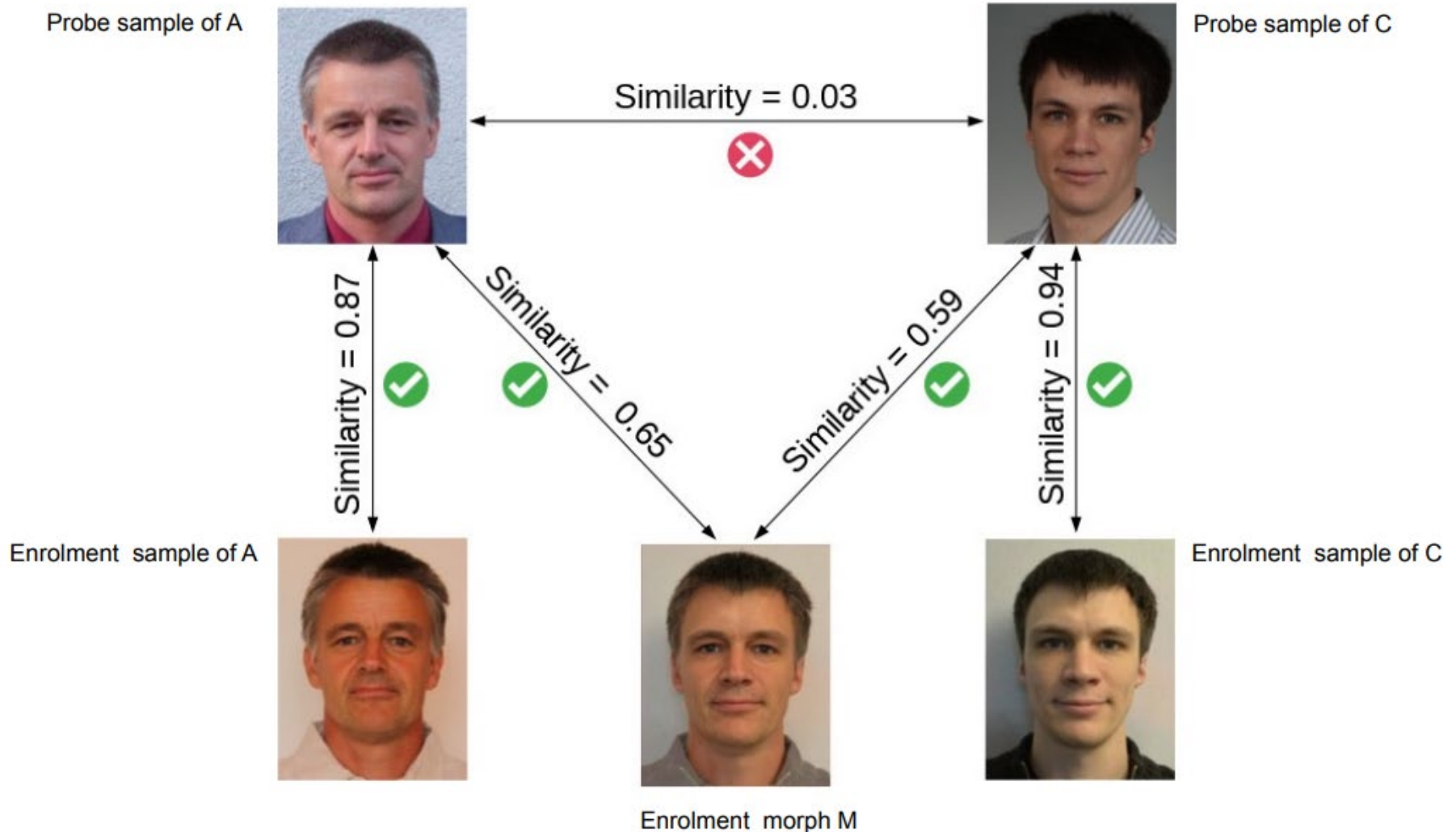
MA: Method – Digital



MA: Method – Printed Photo



Morphing Attacks: Problem



Morphing Attacks: Problem

Multiple individuals, one passport



Morphing Attacks: Problem

In September 2018 German activists

- used a morphed images of Federica Mogherini (High representative of the European Union for Foreign Affairs and Security Policy) and a member of their group
- and received an **authentic German passport**

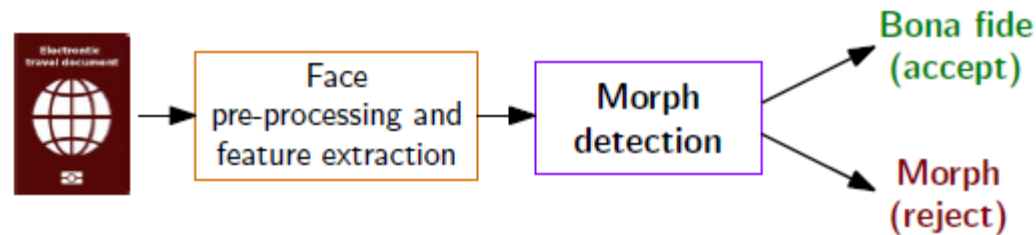


Image source: <https://www.spiegel.de/netzwelt/netzpolitik/biometrie-im-reisepass-peng-kollektiv-schmuggelt-fotomontage-in-ausweis-a-1229418.html>

Morphing Attacks: Scenarios

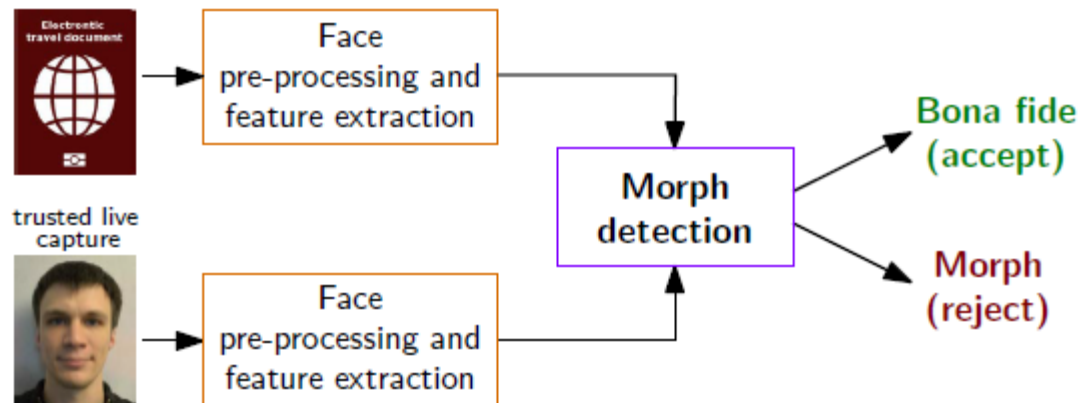
No-reference morph detection

- one **single** suspected image analyzed (e.g. in passport application)



Differential morph detection (e.g. at the border)

- a **pair** of images analyzed, and one is a trusted image (e.g. at the border)



Morphing Attacks: Situation

First investigations on a small dataset

Ongoing evaluations:

- SOTAMD (EU project)
- NIST-FRVT-MORPH (started September 2019)

NIST

PROJECTS/PROGRAMS

<https://www.nist.gov/programs-projects/frvt-morph>

FRVT MORPH

DESCRIPTION

[{Draft Report}](#) [{Participation Agreement}](#) [{API}](#) [{Validation}](#) [{Encryption}](#)

Status

[2019-09-17] The first [draft FRVT MORPH report](#) is now available for public comment. The report includes morph detection performance results for five algorithms over twelve datasets. We welcome feedback and comments to frvt@nist.gov by October 31, 2019.

Overview

Face morphing and the ability to detect it is an area of high interest to a number of photo-credential issuance agencies and those employing face recognition for identity verification. The FRVT MORPH test will provide ongoing independent testing of prototype face morph detection technologies. The evaluation is designed to obtain an assessment on morph detection capability to inform developers and current and prospective end-users, and will evaluate two separate tasks:

1. Algorithmic capability to detect face morphing (morphed/blended faces) in still photographs
2. Face recognition algorithm resistance against morphing

Morphing Attacks: NIST-FATE-MORPH

Update Sep 2021

Single image: **BPCER=1%** -> **APCER 82%**

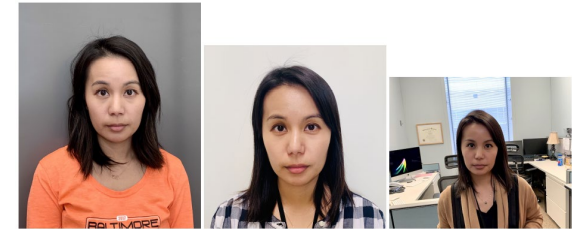
BPCER=10% -> **APCER 20%**

Two images: **BPCER=1%** -> **APCER 94%**

BPCER=10% -> **APCER 8%**

Most recent update (20 June 2023)

- Various software tools, from low quality quality (free) to professional tools



Local Morph Colorized Match (BPCER=0.1)	Local Morph Colorized Match (BPCER=0.01)	Visa-Border (BPCER=0.1)	Visa-Border (BPCER=0.01)	Manual (BPCER=0.1)	Manual (BPCER=0.01)	Print+Scanned (BPCER=0.1)	Print+Scanned (BPCER=0.01)
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BPCER=10%

=1%

=10%

=1%

=10%

=1%

=10%

=1%

Non-morphed images said to be attacks (morphs) -> % of real users annoyed

APCER=3%

=25.9%

=0.7%

=16.1%

=4.9%

=35.7%

=1.2%

=16.9%

Attacks (morphs) said to be a non-morphed/bona fide image -> % of attackers with success

Introduction to Biometric Technologies and Applications (V)

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